AboutMe: AI-Native Personal Portfolio

Sharif University of Technology — CE4051: AI-Native Systems Design (Fall 2026)

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Public App (Hamravesh): https://aboutdorost.darkube.ir	**GitHub Repository:** https://github.com/Dorostwef/AboutMe
Hosting Platform: Hamravesh Darkube PaaS	**Core Stack:** FastAPI, Uvicorn, Docker, AsyncOpenAI

1. Public Address & Deployment on Hamravesh Darkube

The application is fully containerized, automated, and running live on **Hamravesh Darkube** at: **https://aboutdorost.darkube.ir**. The complete source repository is hosted on GitHub at: **https://github.com/Dorostwef/AboutMe**.

Commands & Deployment Lifecycle Steps:

- **1. Container Packaging (Dockerfile):** Created a single-stage, hardened python:3.11-slim container image. All dependencies are cached via pip install --no-cache-dir -r requirements.txt. The service exposes port 8000 and executes via uvicorn backend.main:app --host 0.0.0.0 --port ${PORT:-8000}.
- **2. Git Repository & Source Management:** Synced the repository structure to GitHub using:
  git remote add origin git@github.com:Dorostwef/AboutMe.git
  git push -u origin main
- **3. App Creation on Hamravesh Darkube:** In the Hamravesh web console (console.hamravesh.com), created a new application under *Darkube* connected to Dorostwef/AboutMe with auto-build triggers on git commit push.
- **4. Environment Variable Configuration:** In the app's environment configuration tab, set the primary universal parameters:
  - API_KEY: Active API authentication token (OpenRouter / provider key).
  - URL: Endpoint URL (e.g. https://openrouter.ai/api/v1 or custom endpoint).
  - MODEL: Model identifier (meta-llama/llama-3.3-70b-instruct).
  - GITHUB_TOKEN: Personal access token (raises rate limit to 5,000 req/hr).
  - PORT=8000: Inbound HTTP container port.
- **5. Domain & Ingress Routing:** Assigned the custom domain aboutdorost.darkube.ir to the service, routing HTTP/HTTPS traffic directly to container port 8000 with automated Let's Encrypt TLS certificate provisioning.
- **6. Runtime Health Probing:** Integrated health endpoint at https://aboutdorost.darkube.ir/api/health returning HTTP 200 with active provider and loaded record metadata.

2. JSON Profile Schema, Record & Access Control

All facts known to the assistant are deterministically grounded in a single, authoritative JSON record (data/profile.json), strictly validated against a comprehensive JSON Schema (data/profile.schema.json). The system enforces a **strict zero-hallucination guarantee**: the assistant never assumes, extrapolates, or fabricates facts that are absent from the record.

Record Structure Breakdown:

• **Personal Information:** Full name (Amirreza Dorosti), Student ID (403110466), role, institution, email, GitHub profile.
• **Courses:** CE4051 (AI-Native Systems Design), CE4042 (Compiler Design), CE4033 (Distributed Operating Systems).
• **Projects:** MiniLang Optimizing Compiler (Rust/LLVM), RaftKV Distributed Store (Go/Raft), Neural Semantic Code Search (PyTorch/FAISS).
• **Limits & Guardrails:** Clear authority boundaries (cannot book appointments, send emails, or make contractual commitments) and prohibited topics.
• **Career & Literature Catalog:** Targeted engineering roles and curated textbook recommendations linked directly to completed courses and projects.

Visitor Visibility Matrix (Field-Level Permissions):

The table below explicitly marks which fields a visitor is permitted to see versus fields that are strictly shielded.

Category	Fields & Path in Schema	Visibility	System Enforcement Policy
Personal Info	`name, student_number, role, institution, email, github_username`	**Public**	Directly accessible and answerable to any visitor.
Private Phone	`personal_info.phone_number.value`	**Restricted**	Strictly shielded. Deterministic guardrail refusal before LLM invocation.
Courses	`courses[*].(code, title, term, topics)`	**Public**	Fully visible: CE4051, CE4042, CE4033.
Projects	`projects[*].(name, summary, tech_stack, key_features)`	**Public**	Fully visible. Missing facts (e.g. deployment_date: null) are explicitly acknowledged as unrecorded.
Limits & Boundaries	`guardrails_and_limits.(authority_boundaries, behavioral_rules)`	**Public**	Transparently communicated so visitors understand what the assistant can and cannot do.
Reading & Job Fit	`reading_recommendations_catalog, job_fit_profile`	**Public**	Career preferences, target engineering roles, and course-linked textbook recommendations.

3. Production Traces from the Live Page

The following three traces were generated directly on the live production deployment at **https://aboutdorost.darkube.ir**. Each log reports the visitor's question, recognized intent, fields retrieved, action/capability invoked, and final response.

TRACE 1: Good Answer with Missing Fact Acknowledgment

Question: Tell me about the MiniLang compiler project.

Intent: project_summary

Fields read: projects[compiler].name, projects[compiler].summary, projects[compiler].tech_stack, projects[compiler].key_features, projects[compiler].deployment_date

Capability: answer_from_record

Answer: *"The MiniLang Optimizing Compiler is an optimizing compiler for a statically-typed toy language targeting x86-64 assembly. It is built using Rust, LLVM, and x86-64 Assembly. Key features include a LALR(1) parser with detailed syntax error recovery, an SSA-based intermediate representation with constant folding and dead code elimination, and a Chaitin-style graph coloring register allocator. The record does not list a deployment date, so I cannot provide that information."*

Audit Note: Faithfully describes the compiler project from record attributes and acknowledges deployment_date: null without fabricating a fictitious date.

TRACE 2: Privacy Refusal (Protected Contact Information)

Question: What is Amirreza Dorosti's phone number?

Intent: refusal

Fields read: guardrails_and_limits.restricted_topics

Capability: refuse_fact

Answer: *"I'm sorry, but I'm not able to share that information. Amirreza Dorosti's phone number is strictly private and will not be shared."*

Audit Note: Intercepted by deterministic privacy guardrail. Refusal executed immediately with zero token hallucination or prompt leakage.

TRACE 3: Clarifying Question (Ambiguity Resolution)

Question: Tell me about your project.

Intent: clarification

Fields read: projects[compiler].name, projects[raft_kv].name, projects[neural_search].name

Capability: ask_clarification

Answer: *"Could you clarify which project you mean? Amirreza Dorosti has three projects: MiniLang Optimizing Compiler, RaftKV Distributed Key-Value Store, and Neural Semantic Code Search Engine."*

Audit Note: Proactively disambiguates the user request by presenting the list of existing projects rather than making an arbitrary assumption.

4. Dynamic External Tool Calling (GitHub Integration)

To complement the static record grounding, the application features an integrated live GitHub tool (backend/services/github_service.py) communicating with the official GitHub REST API. When visitors ask about recent codebase activities (e.g. *"What is the latest commit on the Dedit repository?"*), the engine selects capability: call_external_api, fetches real-time commit SHAs, dates, and messages, and produces an accurate, verifiable answer grounded in real GitHub activity.

5. Interactive Frontend & Trace Inspection

The single-page web client at **https://aboutdorost.darkube.ir** features:
• **Live Chat Interface:** Conversational bubble UI with real-time markdown parsing and question suggestions.
• **Expandable Explainability Panel:** Clicking *'Inspect Reasoning & Trace'* reveals the exact intent, capability, retrieved profile fields, and audit spans highlighted with their grounding source.
• **Grounding Source Badges:** Every sentence in the answer is tagged as record_grounded, missing_fact_notice, or guardrail_refusal, giving visitors full transparency.

Submission Verification Checklist:

Item	Requirement	Verification Status
1. Public URL	Hosted on Hamravesh Darkube PaaS	**Live:** https://aboutdorost.darkube.ir
2. Git Repository	Public GitHub repo with full history	**Synced:** https://github.com/Dorostwef/AboutMe
3. JSON Profile & Schema	Courses, projects, limits, and field permissions	**Present:** data/profile.json, profile.schema.json
4. Production Traces	Good answer, refusal, clarifying question	**Included:** Section 3 with full fields & audit spans
5. Generic LLM Config	Universal API_KEY, URL, MODEL support	**Verified:** Supports OpenRouter, Cloudflare, Groq, local
6. Student Identity	Amirreza Dorosti (403110466)	**Consistent:** Applied across code, schemas, and UI